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Summary

- Public environmental preparation for a Hong Kong temperature–cardiovascular admissions study covering January 2013–December 2023 is complete at Hong Kong Observatory (HKO) Headquarters.
- The monthly exposure file contains mean temperature, mean $T_{\min}$, mean $T_{\max}$, and official extreme-day counts for $N = 132$ months. Annual extreme totals match HKO *Year's Weather* summaries for hot nights, very hot days, and cold days across all 11 years (33/33).
- Census and Statistics Department mid-year age–sex population series support person-time denominators, including separate bands for ages 65–69 and 70–74.
- Hospital Authority outcome records are not yet available for analysis. Confirmed extract features include ICD-9 coding, patient-level complete admissions, separate older age bands, and exclusion of ED-to-inpatient episodes from the current file.
- Near-term analytic focus: merge monthly $T_{\max}/T_{\min}$ (and lag temperature) with AMI and stroke admission series; complete Table 1; fit associations only after outcome definition and descriptive quality control.

Prepared materials and current boundary

Prepared

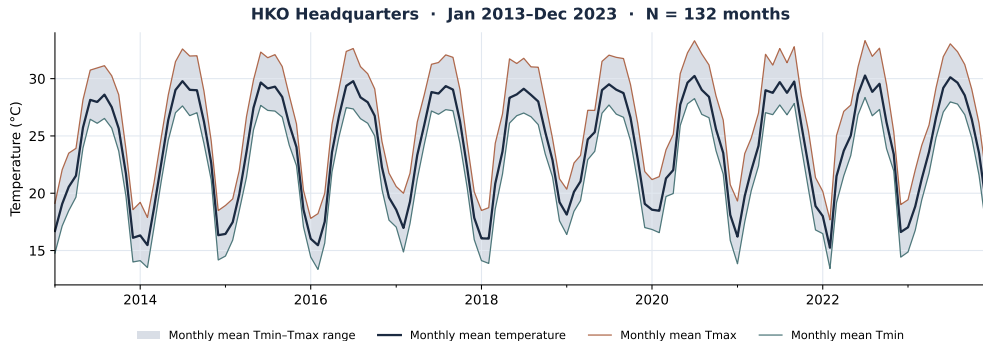
- HKO Headquarters monthly exposure file, 2013–2023
- Extreme-day and spell metrics; absolute humidity
- Validated annual extremes figure and context note
- C&SD age–sex denominators for rate offsets
- Critical literature map and reproducible analysis scripts

Not yet available for inference

- Hospital Authority admissions on the analysis account
- Confirmed medication and BMI field timing
- Real pollution series (placeholder only)
- Influenza series for cold-season sensitivity
- Any temperature–admission coefficient from real outcomes

Boundary. Environmental descriptives may be shown. Substantive admission associations remain pending secure access and schema confirmation.

Monthly temperature, HKO Headquarters, 2013–2023



Environmental exposure only. No admission outcomes.

Table 1 shell — environmental panel complete

Panel A — Environmental observations

Unit: calendar month $N = 132$ HKO Headquarters

Variable	Mean	SD
Mean temperature (°C)	24.02	4.70
Mean Tmin (°C)	22.10	4.69
Mean Tmax (°C)	26.66	4.82
Hot nights (days/month)	3.40	5.40
Very hot days (days/month)	3.19	5.18
Cold days (days/month)	1.10	2.55

Panel B — Admission covariates

Pending Hospital Authority schema and access.

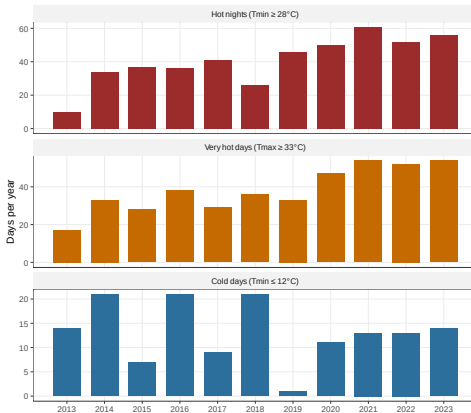
- Age and sex
- Diuretics; beta blockers
- Metformin; SGLT2 inhibitors
- BMI and missingness

Model role depends on the at-risk population and whether covariates are measured before the event window.

Thermal extremes coexist; cold has not disappeared

Official thermal extremes at HKO Headquarters, 2013–2023

Annual counts. Pipeline totals match HKO Year's Weather summaries (33/33).



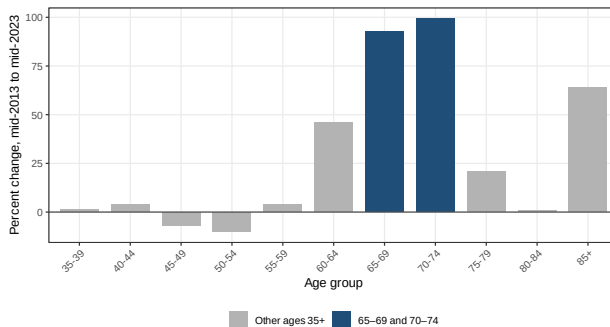
Source: Hong Kong Observatory dailyExtract, Headquarters station. Environmental descriptors only — not health-outcome result

Example contrast: 2013 — 10 hot nights, 14 cold days; 2021 — 61 hot nights, 13 cold days. Annual totals validated 33/33 against HKO *Year's Weather*.

Population aging concentrates person-time in older bands

Mid-year population change by age group, Hong Kong 2013–2023

Percent change in C&SD mid-year population (Table 110-01001). Highlighted bands are intervention-re



Source: Census and Statistics Department, Table 110-01001 (both sexes). Exact mid-year counts, not monthly interpolations.

Mid-year C&SD series: ages 65–69 rose from 295 100 (2013) to 568 500 (2023); ages 70–74 from 213 100 to 424 800. Crude admission counts without age-specific denominators will confound aging with climate.

Literature anchors that constrain claims

- **Goggins et al. (2013).** Hong Kong AMI admissions (2000–2009), daily: cooling below $\sim 24^{\circ}\text{C}$ associated with higher AMI ($\sim 3.7\%$ per 1°C , lags 0–13); heat not the dominant finding; NO_2 relevant.
- **Goggins et al. (2012).** Hemorrhagic and ischemic stroke differ: hemorrhagic stroke shows a strong inverse temperature association (lags 0–4); ischemic stroke weaker and mainly below $\sim 22^{\circ}\text{C}$ (lags 0–13). Subtype separation is required.
- **Guo et al. (2024).** Daily hot-season hospitalizations: official binary hot nights ($T_{\min} \geq 28^{\circ}\text{C}$) not associated with excess risk overall; nighttime heat *intensity* (hot-night excess) was, including circulatory admissions. Binary official counts cannot be treated as self-justifying biology.
- **Wang et al. (2019).** Consecutive extremely hot weather events (including multi-day hot nights and combined day–night patterns) raise mortality risk; event structure matters beyond raw monthly totals.
- **Guo et al. (2025); Yang et al. (2025).** Pollution can modify temperature–hospitalization associations; cold and influenza jointly relate to stroke admissions at weekly scale.

Implications for design and interpretation

- **Trigger versus burden.** Daily distributed-lag models estimate acute triggering. Monthly models estimate hospital-burden associations. Coefficients are not interchangeable with historical per-°C daily estimates.
- **Cold remains co-primary.** A heat-only narrative is inconsistent with local historical evidence and with continued cold-day occurrence in high-heat years.
- **Official extremes versus intensity.** Monthly hot-night counts remain policy-aligned; spell and intensity-style alternatives are scientific checks. A null official-count association would be compatible with Guo et al. (2024).
- **Novelty discipline.** Claims of being the first Hong Kong temperature–CVD or hot-night hospitalization study are not defensible.
- **Near-term exposure emphasis.** Supervisory guidance directs the first merge toward monthly $T_{\max}/T_{\min}$ and lag temperature for AMI and stroke admission risk, with extreme-day metrics retained as secondary specifications.

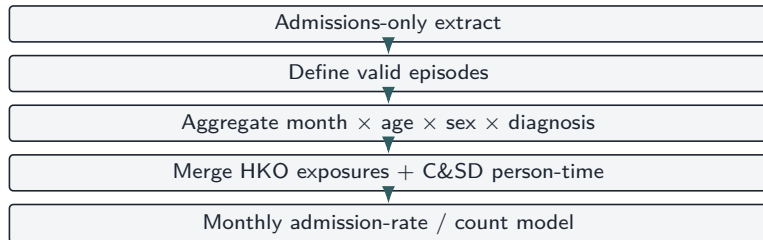
Working hypotheses (prespecified; untested on real outcomes)

1. Colder monthly conditions, and higher cold-day counts, associate with higher AMI and ischemic-stroke admission rates.
2. Higher monthly $T_{\max}/T_{\min}$, or heat-extreme burden, may associate with admissions, especially in older age bands; null findings for official binary hot nights remain scientifically interpretable.
3. Associations differ across AMI, ischemic stroke, and hemorrhagic stroke.
4. Age and clinical covariates (medications, BMI), when measured with valid pre-event timing, may modify temperature–admission associations.
5. Pollution and influenza adjustments are staged sensitivities, not prerequisites for the first descriptive Hospital Authority report.

Analysis unit: one schema fork

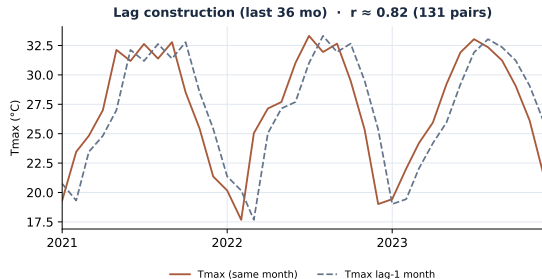
Does the extract contain admissions only, or a defined cohort with non-event follow-up?

Default for the first analysis (admissions-only)



Alternative only if non-event follow-up exists: patient-month risk model. Until the denominator is confirmed, language should refer to monthly admission counts or rates, not patient-level risk.

Temperature variables and lag sequence



- Analysis file can carry same-month and lag-1 mean, Tmin, and Tmax.
- Same-month versus lag-1 mean Tmax:
 $r = 0.8225$ ($n = 131$ pairs).
- Fit temperature specifications separately; avoid one saturated model.
- Proposed sequence: same-month initial; lag-1 sensitivity; mean of current and previous month as secondary.
- Extend processed weather through December 2012 so January 2013 has valid lag-1.

Confirmations required before the first outcome report

1. Meaning of “DAE records included” in the ICD-9 extract.
2. Row/episode unit; principal versus any-position diagnosis; transfer and recurrence rules.
3. Whether the extract is admissions-only or a cohort with non-event months.
4. Presence and timing of diuretics, beta blockers, metformin, SGLT2 inhibitors, and BMI.
5. Primary temperature metric for the first model (mean, T_{min}/T_{max}, or extreme-day counts as secondary).
6. Access pathway timing (authorization signatures and secure-server onboarding).
7. Whether an ethics-protocol amendment is required for this Hospital Authority use.

Next deliverable after access and schema confirmation: validated data dictionary and quality-control report; descriptive admission counts and age-specific rates; completed Table 1. Regression begins only after outcome definition and descriptive quality control are approved.